Architecture Specification & Engineering Reference (Academic Prototype)

AI-Driven Social Media Audience Intelligence Framework

Academic Research Prototype | 7th Semester CSE Minor Project | Team of 4 (P1–P4)

Reference Hardware: NVIDIA GTX 1650 Ti (4GB VRAM) | 16GB RAM | 1TB SSD | Offline-First (v4.8)

> **Document Status:** Canonical Living Architecture Document (v4.8 — Hardware Realism, Sequential VRAM Pipeline & Offline-First Product Rule).
> **System Classification:** Academic Proof-of-Concept (PoC) / Research Testbed. **NTRO is the public problem-statement source only** — this is not an operational intelligence deployment, is not deployed by any defense agency, and does not claim operational intelligence capability.
> **Audience:** Core Engineering Team (P1–P4), Faculty Reviewers, and Viva Defense Examiners.

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1. System Overview & Problem Statement

This system is a **student academic prototype** that monitors **30 curated public Telegram broadcast channels** (and associated public discussion threads), runs five analytical agents (A–E), and surfaces findings on an interactive Bento Grid dashboard plus a grounded retrieval interface over **agent summaries**.

It is motivated by public audience-intelligence problem statements (including an NTRO-originated public statement used for the minor project). It does **not** claim NTRO operational capability or enterprise SOCMINT status.

The Cardinal Product Rule (Offline-First Invariant)

"Build the offline demo as the primary product; treat live Telegram collection and cloud LLMs as optional depth."
Campus Wi-Fi drops, Telegram FloodWait rate limits, and missing API keys are fatal viva failure modes. The primary deliverable is a deterministic, air-gapped seed data replay script (`backfill.py --offline` / SQL seed) and local LLM inference via Ollama (`llama3.2:3b`). Live Pyrofork MTProto ingestion is an optional depth feature demonstrated only if the campus network allows.

Core Analytical Capabilities:

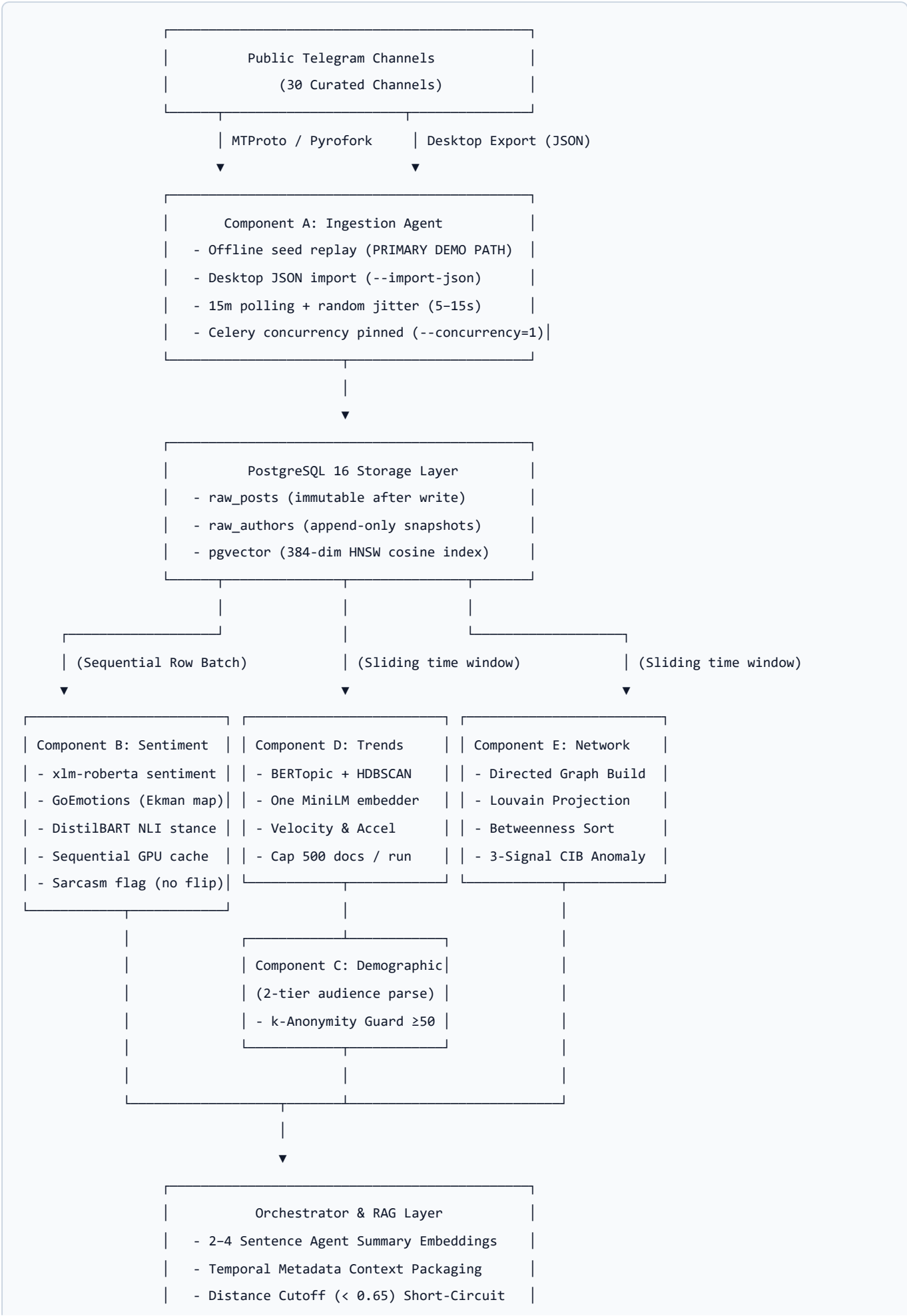
1. **Longitudinal Sentiment & Stance Trajectories:** Measuring population-level emotional resonance and policy stance over time with off-the-shelf pre-trained transformers, without conflating emotion with ideological alignment.
2. **Emergent Topic Velocity & Acceleration:** Identifying accelerating narratives before volume peaks, distinguishing organic interest from coordinated amplification.

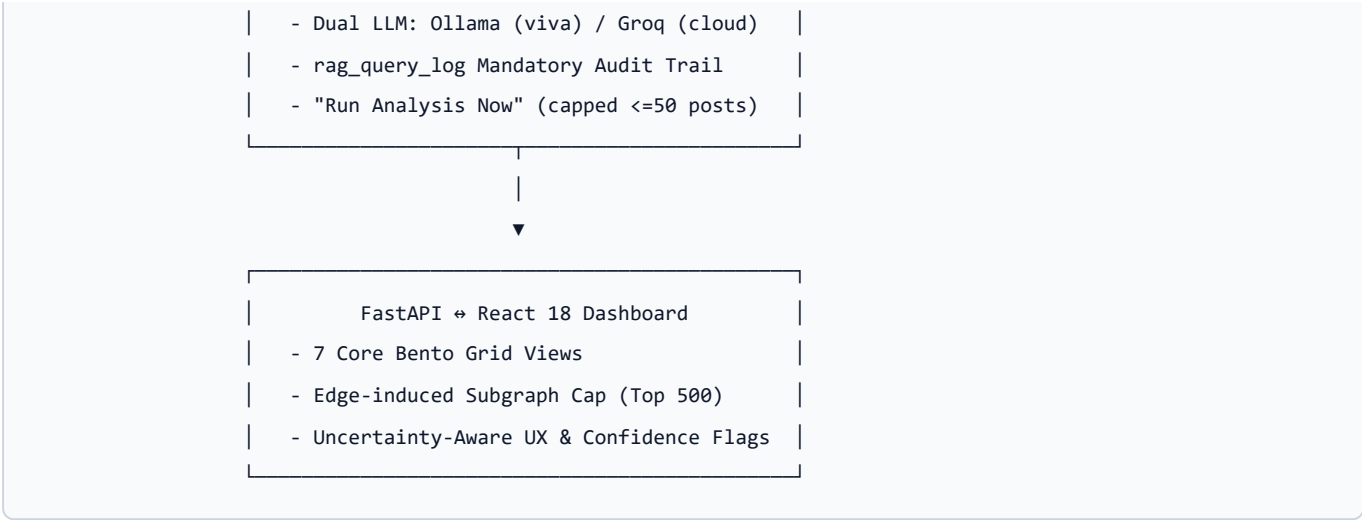
3. **Coordinated Inauthentic Behavior (CIB) Screening:** Detecting topological and temporal synchrony across broadcasting entities and active participant clusters using an intuitive 3-signal anomaly heuristic.
4. **Grounded Query over Agent Summaries:** An analyst query terminal that retrieves the system's own 2–4 sentence agent summaries and **refuses** when cosine distance is too high (≥ 0.65) — a disciplined retrieve-and-refuse mechanism, not a hallucination-free oracle.

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2. Multi-Agent Pipeline Topology

The platform rejects monolithic processing in favor of **five independently invocable analytical agents (A–E)** coordinated by a lightweight orchestrator, backed by PostgreSQL 16 + pgvector and Redis.


Public Telegram Channels
(30 Curated Channels)
MTProto / Pyrofork
Desktop Export (JSON)
Component A: Ingestion Agent
- Offline seed replay (PRIMARY DEMO PATH)
- Desktop JSON import (--import-json)
- 15m polling + random jitter (5–15s)
- Celery concurrency pinned (--concurrency=1)
PostgreSQL 16 Storage Layer
- raw_posts (immutable after write)
- raw_authors (append-only snapshots)
- pgvector (384-dim HNSW cosine index)
(Sequential Row Batch)
(Sliding time window)
(Sliding time window)
Component B: Sentiment
- xlm-roberta sentiment
- GoEmotions (Ekman map)
- DistilBART NLI stance
- Sequential GPU cache
- Sarcasm flag (no flip)
Component D: Trends
- BERTopic + HDBSCAN
- One MiniLM embedder
- Velocity & Accel
- Cap 500 docs / run
Component E: Network
- Directed Graph Build
- Louvain Projection
- Betweenness Sort
- 3-Signal CIB Anomaly
Component C: Demographic
(2-tier audience parse)
- k-Anonymity Guard ≥50
Orchestrator & RAG Layer
- 2–4 Sentence Agent Summary Embeddings
- Temporal Metadata Context Packaging
- Distance Cutoff (< 0.65) Short-Circuit



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3. Honest Self-Assessment, Hardware Realism & Limitations

In intelligence systems engineering, unstated assumptions and overstated model capabilities are critical vulnerabilities. The following subsections document the explicit mathematical trade-offs, empirical boundaries, and deliberate design compromises embedded in the framework.

3.1 Hardware Budget & Sequential VRAM Pipeline Strategy

- **Target Reference Dev & Demo Machine:**
- **GPU:** NVIDIA GeForce GTX 1650 Ti Mobile (4 GB GDDR6 VRAM)
- **System Memory:** 16 GB DDR4 RAM
- **Storage:** 1 TB NVMe SSD + 1 TB HDD
- **OS:** Windows 11 / WSL2 / Docker Desktop
- **Teammate Minimum Fallback Spec:** 8–16 GB RAM, Quad-Core CPU, CPU-only fallback using quantized models / seed replay.
- **VRAM Memory Footprint Reality Check (4 GB Budget):**
- `cardiffnlp/twitter-xlm-roberta-base-sentiment` : ~1.1 GB
- `SamLowe/roberta-base-go_emotions` : ~500 MB
- `valhalla/distilbart-mnli-12-3` (stance) : ~260 MB (replaces 1.6 GB `bart-large`)
- `paraphrase-multilingual-MiniLM-L12-v2` (embed) : ~470 MB
- Ollama (`llama3.2:3b` 4-bit quantized) : ~2.0 GB
- Postgres + Redis + Celery + FastAPI + React : ~2.5–3.5 GB System RAM
- **Co-resident Total:** >4.3 GB VRAM $\rightarrow$ **Exceeds 4 GB VRAM if loaded concurrently!**
- **Strict Invariant — Sequential GPU Execution Pipeline:**
- To prevent catastrophic CUDA OOM crashes on 4 GB VRAM cards, the worker **never** holds all models resident on the GPU at once.
- Ingestion and preprocessing stage the batch. Component B executes sequentially:
  1. Run XLM-RoBERTa sentiment batch $\rightarrow$ invoke `torch.cuda.empty_cache()`
  2. Run GoEmotions emotion batch $\rightarrow$ invoke `torch.cuda.empty_cache()`
  3. Run DistilBART stance batch $\rightarrow$ invoke `torch.cuda.empty_cache()`

4. Run MiniLM embeddings for BERTopic $\rightarrow$ invoke `torch.cuda.empty_cache()`

- Ollama runs in an independent system process (`ollama serve`). On 4 GB hardware, Ollama automatically shares VRAM or offloads layers to system RAM.
- **Demo Timing Calibration:**
- The "Run Analysis Now" viva trigger finishes in **15–25 seconds on the reference GTX 1650 Ti** ONLY for a capped batch of **≤ 50 seed posts**.
- Historical bulk analysis runs asynchronously in background via Celery Beat, not instantaneously.

3.2 Stance Threshold Calibration & DistilBART Runtime Default

Stance detection in Component B is operationalized using Natural Language Inference (Zero-Shot NLI via `valhalla/distilbart-mnli-12-3`) rather than a supervised classifier, evaluating posts against target hypotheses ("*This text expresses support for {topic}*" vs. "*This text expresses opposition to {topic}*"). - **Runtime Default:** `valhalla/distilbart-mnli-12-3` (~260 MB, 6x smaller and 3x faster than BART-large). - **Evaluated Baseline:** `facebook/bart-large-mnli` (~1.6 GB) is evaluated offline in the report for benchmark comparison, but is excluded from the default live runtime to protect the 4 GB VRAM budget. - **Threshold Calibration:** The classification boundary enforces a hard threshold of **0.60**: if neither hypothesis attains ≥ 0.60 entailment probability, or if contradictory signals emerge, the pipeline assigns `'neutral'` or `'unclear'` rather than forcing a binary categorization. - Setting the decision boundary at 0.60 explicitly trades recall for precision: weakly committed posts fall into `'neutral'`, ensuring that posts labeled `'supportive'` or `'against'` possess high-confidence entailment evidence.

3.3 Sarcasm Detection Ceiling & Advisory Uncertainty Bounds

Linguistic, syntactic, and emoji-based sarcasm heuristics (in `sentiment_b/sarcasm.py` via valence-emoji incongruity, intensifier contradiction, and punctuation clustering) have an empirical F1 ceiling typically reported between **50% and 65% in the literature** (e.g., SemEval-2018 Task 3 irony baselines; SARC 2.0 benchmarks; Swami et al., 2018). In South Asian multi-dialectal and code-mixed environments (English + Romanized Hinglish), sarcasm depends heavily on unstated cultural context.

Acknowledging this empirical ceiling, the framework treats sarcasm detection **strictly as an advisory uncertainty flag (`sarcasm_flag = TRUE`) and confidence penalizer, never as an automated valence inverter**. The pipeline never flips a positive classification to negative simply because a sarcastic emoji was detected. Flagged posts are marked with a visible "Lower Confidence / Sarcasm Flagged" warning in the UI, preserving aggregate transparency without introducing compounding error.

3.4 Operational Security Posture & Threat Model Demarcation

The security architecture is designed and bounded as an **academic Proof-of-Concept (PoC) and evaluation testbed**, optimized for developer ergonomics, multi-agent orchestration demonstration, and local reproducibility during viva defense. Access control relies on HTTP Basic Authentication over FastAPI, local `.env` configuration, internal Docker bridge networks, and strict SQLite session single-process concurrency controls (`--concurrency=1` on Celery) to prevent MTProto `.session` file corruption.

This architecture explicitly **does not represent a hardened, production-grade SOCMINT deployment**. Sovereign intelligence systems require mutual TLS (mTLS), Hardware Security Modules (HSM), fine-grained RBAC/ABAC, and air-gapped egress proxies.

3.5 3-Signal Arithmetic Mean for CIB Screening

Rather than asserting unvalidated, pseudo-precise statistical regression weights across an unannotated Telegram domain, Component E composites three transparent, intuitive heuristic sub-signals into an unweighted arithmetic mean: $$\text{anomaly_score} = \frac{s_1 + s_2 + s_3}{N_{\text{available}}}$$ where: - s_1: **Posting Synchrony** — accounts posting within unusually tight time windows relative to one another (using platform `posted_at`). - s_2: **Account Age vs. Volume** — young accounts generating disproportionately high message volumes. - s_3: **Content Duplication** — near-duplicate message text via TF-IDF cosine similarity.

If $\text{anomaly_score} \ge 0.70$, the entity is flagged with `coordination_signal = TRUE`. This formulation is robust, transparent to examiners, dynamically averages over available signals, and avoids inventing complex weight parameters that have no ground-truth empirical calibration on Telegram data. It is an **advisory screening indicator**, never a legal verdict.

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4. Component Technical Specifications

Component A: Ingestion Agent

- **Primary Product Deliverable (Offline Viva Path):** Standalone replay script (`ingestion_a/backfill.py --offline`) loads historical posts from `scripts/seed_demo_data.sql` with normalized timestamps into PostgreSQL. **The viva must succeed on this path with zero internet connectivity.**
- **Live MTProto Driver:** Pyrofork wrapped in `BaseIngestionAdapter`. Staggered continuous polling: 1 channel polled every 30 seconds with 2–5s randomized jitter across a 15-minute cycle ($30 \times 30\text{s} = 900\text{s}$), maintaining a global rate of ~ 2 requests/min to eliminate Telegram API FloodWait. Celery worker concurrency locked strictly to `--concurrency=1` to prevent SQLite `.session` database lock contention.
- **Telegram Desktop JSON Import:** Script (`ingestion_a/import_exported.py --import-json <path>`) parses Telegram Desktop JSON exports into `raw_posts` with `processing_status = 'pending'`, providing historical depth without API dependencies.

Component B: Sentiment, Emotion & Stance Agent

- **Off-the-Shelf Pre-trained Transformers (No Custom Training Claimed):**
- **Sentiment (all text):** `cardiffnlp/twitter-xlm-roberta-base-sentiment` — multilingual sentiment classifier (positive/negative/neutral), pre-trained on 198M tweets across 8 languages. Handles English and Hinglish (Roman-script code-mixed Hindi-English) without routing.
- **Emotion (English/Hinglish):** `SamLowe/roberta-base-go_emotions` — 28-class emotion classifier mapped to 7 core dimensions (`joy` , `anger` , `sadness` , `disgust` , `surprise` , `neutral` , `fear`).
- **Stance (all text):** Zero-Shot NLI via **`valhalla/distilbart-mnli-12-3`** **(runtime default locked)** against target hypotheses. Hard 0.60 threshold.
- **Sequential GPU Execution:** Enforces `torch.cuda.empty_cache()` between pipeline stages to guarantee execution within 4 GB VRAM.
- **No Devanagari Regex Language Router:** Target channels produce English and Romanized Hinglish content. Multilingual XLM-RoBERTa handles both natively.
- **Sarcasm Flag:** Advisory penalty only (`sarcasm_flag = TRUE`). Never inverts sentiment polarity.

Component C: Demographic Agent

- **Operational Boundary:** Telegram subscriber rosters are strictly private. Profiling operates across two observable heuristic tiers: 1. *Broadcast Tier:* Channel-level language, timezone diurnal posting rhythms, and description keywords. 2. *Audience Tier:* Inferred distributions derived strictly from active participants in linked public discussion groups and comment threads.
- **k-Anonymity Cluster Guard:** Aggregate-only outputs. Any cluster with account count < 50 is flagged `below_threshold: true` and greyed out in the UI. UI title: **Audience aggregates (inferred)** — coarse bio/keyword heuristic, not census demographics.

Component D: Trend & Topic Agent

- **Model:** BERTopic utilizing `paraphrase-multilingual-MiniLM-L12-v2` (384 dimensions, 50+ languages). Single shared embedder for both BERTopic and RAG.
- **Sparse Corpus Tuning:** Calibrated to `min_topic_size = 3` and `topic_model.reduce_outliers(..., strategy="c-tf-idf")`. **Input documents capped at 500 per run** to eliminate memory spikes.
- **Temporal Metric:** Ranks topics by velocity ($v = \frac{\Delta \text{posts}}{\Delta t}$) and acceleration ($a = \frac{\Delta v}{\Delta t}$) calculated strictly using `posted_at` timestamps (never `ingested_at`).

Component E: Network & CIB Agent

- **Graph Construction:** Directed multigraph of interactions (`forward` , `reply` , `mention`) stored in `network_edges_e`.
- **Community Projection:** Directed graph is projected into an undirected representation (`G.to_undirected(as_view=False)`) prior to executing Louvain modularity (`python-louvain` , imported as `import community`).
- **Canvas Protection & Edge Induction:** Centrality sorting prioritizes Betweenness Centrality. Graph payload is capped at top 500 nodes. Edges are strictly filtered to the induced subgraph.
- **3-Signal CIB Screening:** Heuristic average of posting synchrony (s_1), age-to-volume ratio (s_2), and content duplication (s_3). Flagged if score ≥ 0.70.

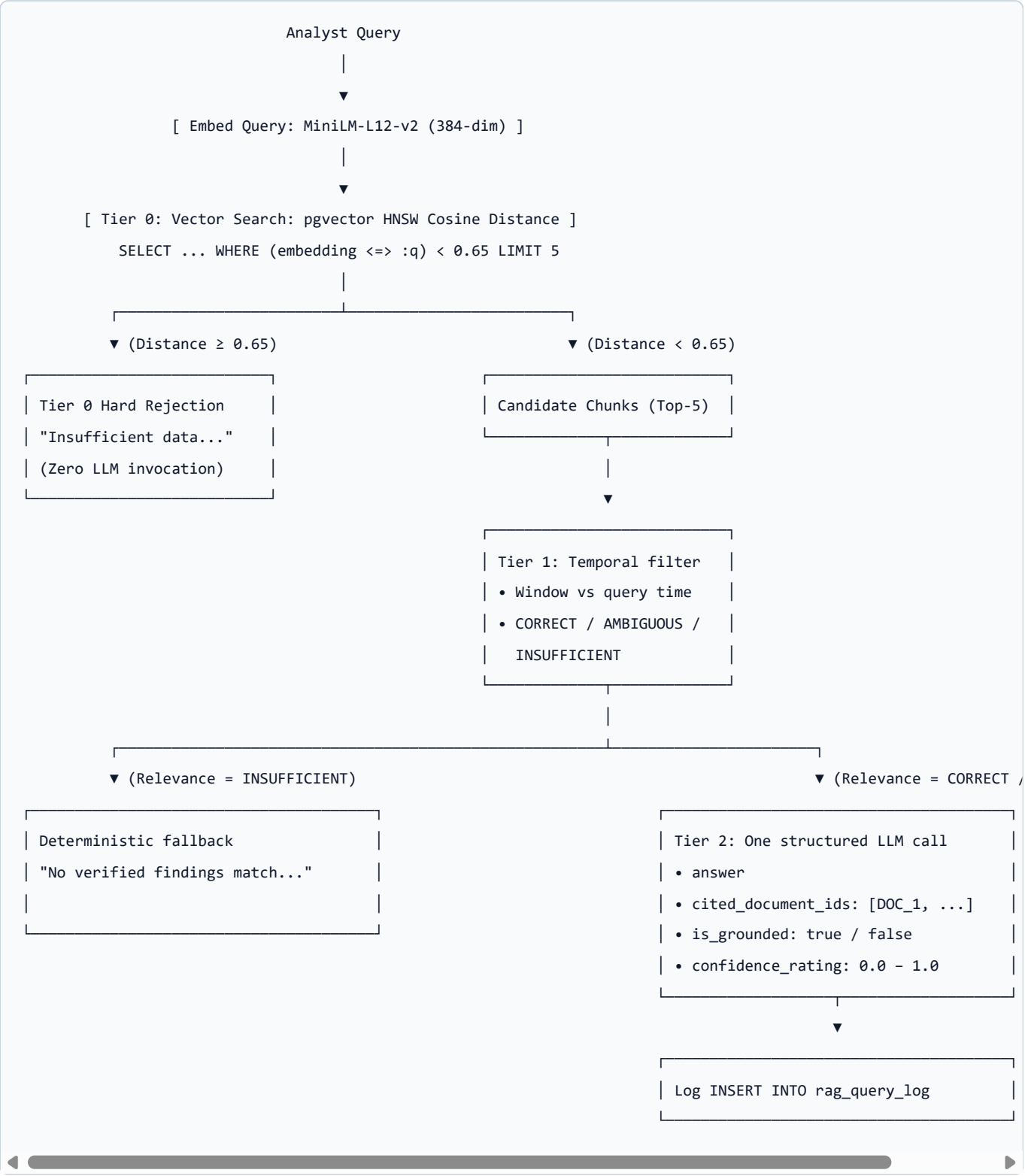
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5. Grounded Retrieval (RAG) — Retrieve, Refuse, One LLM Pass

The query layer retrieves **agent summaries** (not noisy raw posts) and generates an answer with citations. It is a pragmatic retrieve-and-refuse pipeline with a single structured LLM pass, not a published CRAG/Self-RAG implementation or hallucination-free oracle.

- **Offline Viva Default:** Ollama `llama3.2:3b` on `localhost:11434` (zero internet dependency).
- **Optional Development:** Groq `llama-3.3-70b-versatile` when an API key is available. (No unverified "98% grounding accuracy" marketing claims).
- Configured via `LLM_PROVIDER=ollama|groq|openai` in `.env`.

5.1 Pipeline Topology



5.2 Deterministic Grounding & Verification Guarantees

1. **Summary-Level Indexing:** Raw posts are never embedded directly into the vector store. Instead, analytical agents (B, C, D, E) generate dense, structured 2–4 sentence summaries indexed with strict temporal metadata (`window_start` to `window_end` in UTC).
2. **Tier 0 Hard Mathematical Cutoff (`< 0.65`):** Queries matching no summary chunks with cosine distance < 0.65 trigger an immediate short-circuit. The LLM is **never called**, providing an unbreachable mathematical

guarantee against hallucinating on queries outside the monitored domain.

3. **Tier 1 Temporal Relevance Filtering:** Candidate documents passing Tier 0 are filtered for temporal consistency against query timeframe envelopes.
4. **Tier 2 Single-Pass Structured Contract:** The prompt enforces a Pydantic structured output contract (`GroundedAnalystResponse`): `relevance_grade` , `answer` , `cited_document_ids` , `is_grounded` , and `confidence_rating` . Single pass avoids multi-turn loops that would freeze local Ollama inference on a laptop.
5. **Tier 3 Audit Trail Logging:** Non-bypassable logging to `rag_query_log` occurs prior to returning the HTTP payload, recording query text, retrieved chunk IDs, generated response, confidence rating, and grounding status.

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6. Data Governance, Regulatory Alignment & Post-Viva PII Scrub

6.1 Gated Research Authorisation & OSINT Boundary

- **Public OSINT Boundary:** Ingestion operates strictly on publicly accessible Telegram broadcast channels via Pyrofork MTProto. No private groups, private chats, or encrypted direct messages are ever accessed.
- **Supervisor Gating:** Operational channel monitoring is formally signed off with the project faculty supervisor as research into public social communication spaces.

6.2 Regulatory Alignment (CERT-In Directions & DPDP Act 2023)

While the framework is an academic research prototype exempt from commercial intermediary reporting obligations, the system aligns with core principles: 1. **CERT-In Timestamp Synchronization:** All ingested events and analytical timelines are strictly anchored to UTC `TIMESTAMPTZ` (`posted_at`), ensuring non-skewed chronological ordering. 2. **CERT-In Analyst Auditability:** Every analyst query against the RAG interface is logged with query text, timestamp, and retrieved document IDs into `rag_query_log` . 3. **DPDP Act 2023 Compliance via k-Anonymity ($k \ge 50$):** In Component C, individual-level profiling is prevented. Demographic clusters with fewer than 50 accounts are suppressed (`below_threshold: true`). 4. **Advisory Anomaly Detection:** Component E outputs anomaly scores as probabilistic screening signals (0.0 to 1.0), never definitive verdicts. 5. **Protective Pseudonymization:** Monitored authors in graph visualizations are masked (e.g., `Actor-[Masked]-XXX`).

6.3 Post-Viva PII Disposal Routine

To satisfy ethical data retention without building complex custom database infrastructure, personal text and user metadata are wiped from the development database post-viva:

```
-- Committed End-of-Semester PII Scrub Routine (scripts/post_viva_scrub.sql)
BEGIN;

-- 1. Wipe raw message content and media attachments
UPDATE raw_posts
SET content_text = NULL,
    media_urls = NULL;

-- 2. Wipe author personal identifiers
UPDATE raw_authors
SET display_name = NULL,
    bio_text = NULL;

COMMIT;
```

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7. Storage Performance & Indexing Scheme

```
-- 1. Component B row-level processing queue (high-frequency worker poll)
CREATE INDEX idx_raw_posts_pending ON raw_posts (id) WHERE processing_status = 'pending';

-- 2. Temporal range queries for sliding windows (Components C, D, E)
CREATE INDEX idx_raw_posts_posted_at ON raw_posts (posted_at DESC);

-- 3. Vector cosine distance index via HNSW
CREATE INDEX idx_rag_documents_hnsw ON rag_documents USING hnsw (embedding vector_cosine_ops);
```

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